

the 1960 median. These shifts are visible from 1960 to 1980 and become even more pronounced by 1998. Adding variation along the intensive margin augments the rightward shift, particularly for the nonroutine interactive measure.

Panels C and D of Figure II plot the corresponding densities for routine cognitive and routine manual task input. Consistent with the theoretical model, the distribution of labor input of both routine cognitive and routine manual tasks shifted sharply leftward after 1960—opposite to the case for nonroutine tasks. The shift is particularly pronounced for the routine cognitive task measure, and becomes even more apparent when the intensive margin is added. As suggested by Figure I, the decline in the input of routine manual tasks is less dramatic, though still visible.

In sum, the evidence in Figures I and II supports our model's primary macroeconomic implications. Between 1970 and 1998 there were secular declines in labor input of routine cognitive and routine manual tasks and corresponding increases in labor input of nonroutine analytic and interactive tasks. We next analyze the sources of these task shifts at the industry level.

III.B. Task Changes within and between Industries

The changes in economywide labor input of routine and non-routine tasks documented in Figure I and Table II could stem from substitution of computer capital for routine labor inputs within detailed industries, as our model suggests. Alternatively, they could stem from changes in the composition of final demand. Since much of our detailed analysis focuses on changes in task input at the industry level, we explore briefly the extent to which changes in job content are due to within-industry task shifts.

Panel B of Table II presents a standard decomposition of task changes into within- and between-industry components.¹⁶ This decomposition shows quite consistent patterns of task change.

16. We decompose the use of task k in aggregate employment between years t and τ ($\Delta T_{k\tau} = T_{k\tau} - T_{kt}$) into a term reflecting the reallocation of employment across sectors and a term reflecting changes in task j input within industries using the equation $\Delta T_{k\tau} = \sum_j (\Delta E_{j\tau} \gamma_{jk}) + \sum_j (\Delta \gamma_{jk\tau} E_j) = \Delta T_{k\tau}^b + \Delta T_{k\tau}^w$, where j indexes industries, $E_{jk\tau}$ is the employment of workers in task k in industry j in year τ as a share of aggregate employment in year τ , E_{jt} is total employment (in FTEs) in industry j in year τ , $\gamma_{jk\tau}$ is the mean of task k in industry j in year τ , $\gamma_{jk} = (\gamma_{jk\tau} + \gamma_{jk t})/2$, and $E_j = (E_{j\tau} + E_{jt})/2$. The first term ($\Delta T_{k\tau}^b$) reflects the change in aggregate employment of task k attributable to changes in employment shares between industries that utilize different intensities of task k . The second term ($\Delta T_{k\tau}^w$) reflects within-industry task change.

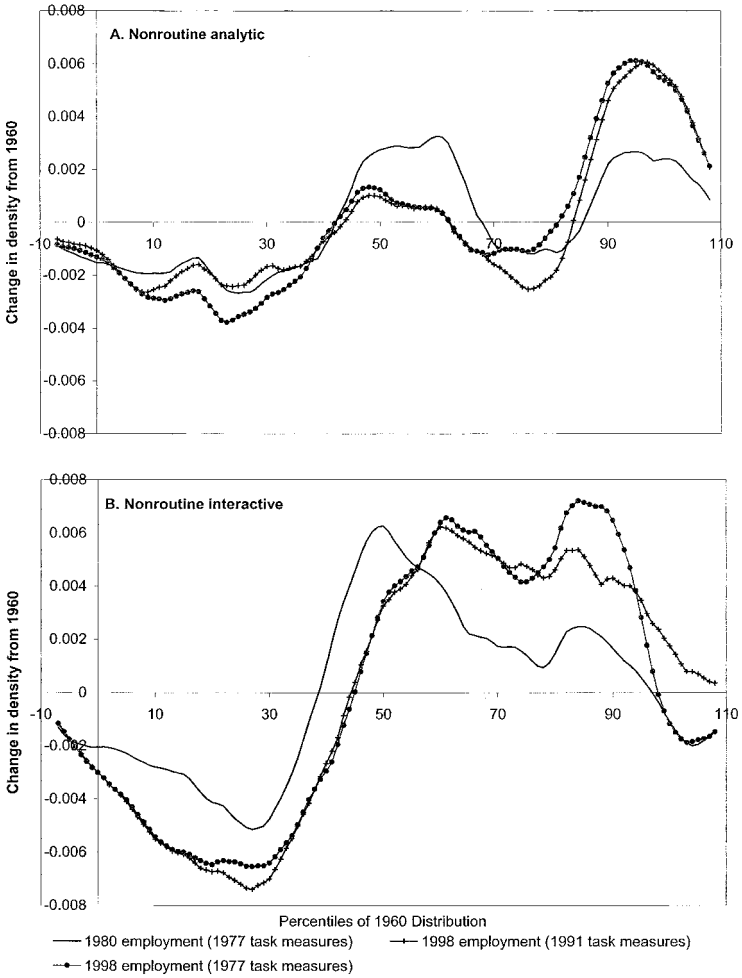


FIGURE II
Smoothed Differences between the Density of Nonroutine Task Input
in 1960 and Subsequent Years

Figure II is constructed using *Dictionary of Occupational Titles* (DOT) task measures by gender and occupation paired to employment data from 1960, 1980, and 1998 Census and Current Population Survey samples. Plots depict the change in the share of employment between 1960 and the indicated year at each 1960 percentile of task input. All series use DOT 1977 data paired to employment data for the indicated year except for series marked “1991 task measures,” which use task data from 1991 DOT. See Table I and Appendix 1 for definitions and examples of task variables.

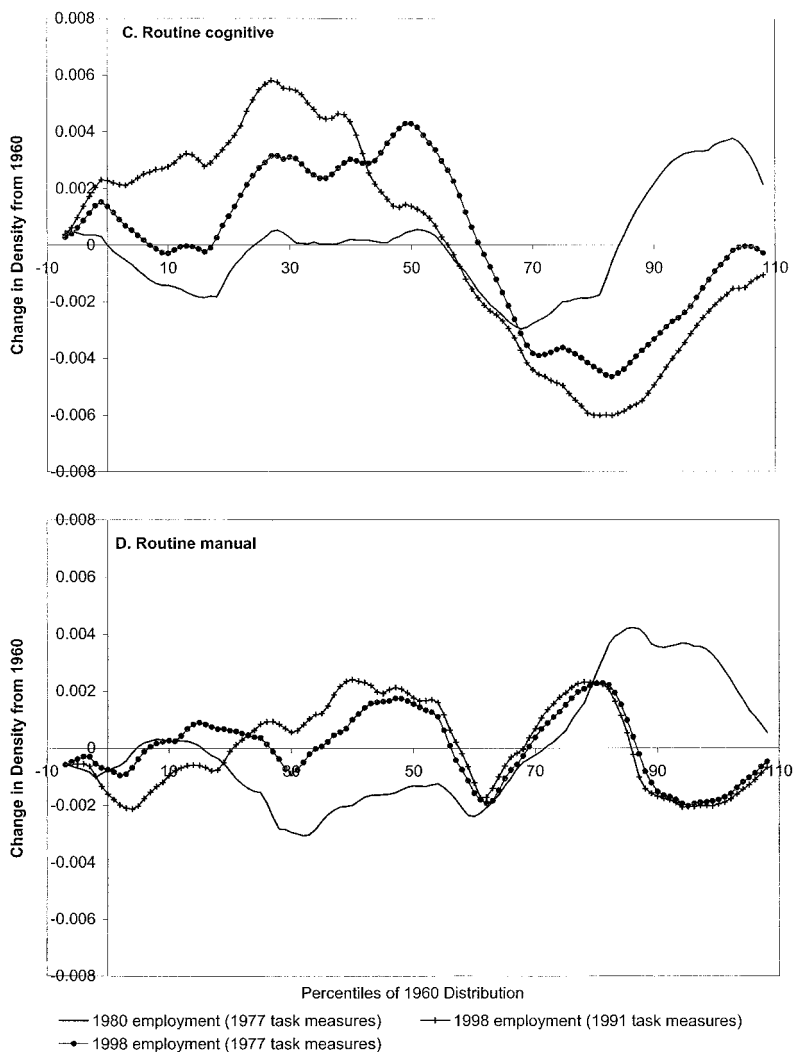


FIGURE II (CONTINUED)

Both the nonroutine analytic and nonroutine interactive task measures show strong within-industry growth in each decade following the 1960s. Moreover, the rate of within-industry growth of each input increases in each subsequent decade. Although, as noted above, nonroutine analytic input also increased during the 1960s, Table II shows that this was primarily a cross-industry

phenomenon—i.e., due to sectoral shifts. After the 1960s, by contrast, the growth in nonroutine task input was dominated by within-industry task shifts.

Trends in both routine cognitive and routine manual tasks show a similarly striking pattern. Both types of routine task input increased during the 1960s, due to a combination of between- and within-industry shifts. In the decades following, however, input of both routine cognitive and routine manual tasks sharply declined, and the bulk of these declines was due to within-industry shifts. Moreover, the rate of within-industry decline increased in each subsequent decade.

As distinct from the other four task measures, we observe steady within- and between-industry shifts against nonroutine manual tasks for the entire four decades of our sample. Since our conceptual framework indicates that nonroutine manual tasks are largely orthogonal to computerization, we view this pattern as neither supportive nor at odds with our model.

In summary, the trends against routine cognitive and manual tasks and favoring nonroutine cognitive tasks that we seek to analyze are dominated by within-industry shifts, particularly from the 1970s forward. We next analyze whether computerization can explain these task shifts.¹⁷ Because our model makes no prediction for how computerizing industries will adjust demand for nonroutine manual tasks, we do not include this variable in our industry-level analysis below (see Autor, Levy, and Murnane [2001] for detailed analysis).

IV. COMPUTERIZATION AND TASK CHANGE: INDUSTRY LEVEL RELATIONSHIPS

As industries adopt computer technology, our model predicts that they will simultaneously reduce labor input of routine cognitive and manual tasks and increase labor input of nonroutine cognitive tasks. We test these hypotheses below.

17. Our model also implies that the expenditure shares of routine-task-intensive industries should have increased as ρ declined. By contrast, the prediction for the employment share of routine-task-intensive industries is ambiguous since these industries should have differentially substituted computer capital for labor input. Because our data measure employment, not expenditures, we are unable to test the implication for expenditure shares. Closely related, computer-intensive industries should have experienced relatively larger gains in labor productivity as ρ declined. Stiroh [2002] presents evidence that this occurred in the 1990s.